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LexiEase AI : An AI Powered Dyslexic Support System

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***Abstract* — The development of an intelligent, dyslexia-specific support system is the need of the hour to ensure inclusion and personalization. The work presents an artificial intelligence-driven system, which involves state-of-the-art machine learning and NLP techniques, for identifying and assisting dyslexics. Our system encompasses a broad range of dyslexia screening tests, such as phonological awareness tests, memory tests, and oral reading, for which dyslexics may be tested using state-of-the-art models like DistilBERT and DeepSpeech. A Random Forest classifier generates certain dyslexia score to classify the severity levels and give customized learning pathways. The system will also integrate document simplification, mind map generation, and writing assistance features in order to make it holistic.**

***Keywords* —** *DeepSpeech, DistilBERT, Personalized Learning,, Document Simplification, Mind Map Generation, Writing Assistance.*

I. INTRODUCTION

Dyslexia is a common learning disability that makes it very hard for people to acquire basic literacy skills, hence affecting their performance at school and in everyday life. While most traditional methods of addressing dyslexia focus on generic strategies, the growing need has been for personalized and technology-driven solutions. Recent developments in AI and NLP have opened new avenues for creating tailored support systems to address this gap.

This work proposes a new AI-enabled system for screening, assessing, and supporting dyslexia. Our approach encompasses the integration of state-of-the-art AI models: DistilBERT for semantic analysis and DeepSpeech for speech recognition, thus allowing speedy and accurate assessments of dyslexia. The system shall carry out three major tests to constitute the screening, namely phonological awareness, memory evaluation, and oral reading fluency—all analyzing different aspects of cognitive ability. From these results, based on the Random Forest that gives the degree of dyslexia, suggestions on strategies for learning are made.

The system goes beyond diagnostics to constitute a bundle of tools for learning and engagement, such as document summarization with AI-driven simplification, the generation of mind maps in an interactive way, and a dynamic writing assistant also powered by AI. This will improve both your comprehension and productivity as a dyslexic.

II. LITERATURE REVIEW

Di Fraia, G. [1] The paper introduces DSAi, an AI-powered platform designed to support students with Specific Learning Disorders (SLDs) by enhancing inclusion, accessibility, and innovation in education.

Goodman, S. M., Buehler, E. [2] The research presents LaMPost, an AI-powered email-writing tool using large language models to assist adults with dyslexia by providing features like rewriting, summarizing main ideas, and generating subject lines.

Iyer, L. S., Chakraborty. [3] The paper explores the application of AI-assisted language learning models to support children with dyslexia and dysgraphia, enhancing their reading and writing skills through personalized, adaptive learning experiences.

Goodman, S. M., Buehler.[4] The research paper introduces LaMPost, an AI-assisted email writing prototype designed to support adults with dyslexia by offering features such as outlining main ideas, generating subject lines, and suggesting changes.

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Wu, S., Reynolds. [5] The research paper presents the Additional Writing Help (AWH) tool, designed to assist individuals with dyslexia in writing on social media by providing tailored spelling and grammar suggestions through a neural machine translation model.

Dabaghi, K., D'Urso. [6] The research paper discusses the implementation of AI-based tools to enhance learning for students with dyslexia, focusing on their effectiveness in improving reading and writing skills.

Gilbert, B., Stubblefield, J. [7] The research paper explores the use of artificial intelligence to develop a novel font aimed at improving reading abilities in individuals with dyslexia, based on performance data collected from various fonts.

Alghabban, W. G., Salama. [8] The research paper presents a multimodal interface tool developed on cloud computing technology to support students with dyslexia, utilizing audio, text, and images to enhance learning.

Rodríguez-Cano, S., Delgado-Benito. [9] The research paper details the design of a virtual reality application aimed at improving the educational experiences of students with dyslexia, focusing on user-centered design principles.

Maresca, G., Leonardi, S. [10] The research paper investigates the relationship between dyslexia and visuo-spatial attention, proposing that deficits in visual attention contribute to reading difficulties in dyslexic individuals.

Spoon, K., Siek, K., Crandall, D. [11] The research paper explores the use of AI to detect dyslexia in children's handwriting, demonstrating that machine learning models can identify handwriting features indicative of dyslexia with a notable accuracy of 77.6%.

Alkhurayyif, Y., & Sait, A. R. W. [12] The research paper highlights the effectiveness of models like CNNs and SVMs in classifying dyslexic and non-dyslexic individuals.

Isa, I. S., Rahimi, W. N. S. [13] The research paper examines the use of technology to assist children with dyslexia in reading, focusing on various digital tools and interventions that enhance literacy skills.

Al-Barhamtoshy, H. M., & Motaweh, D. M. [14] The research paper presents a framework for diagnosing dyslexia using computational analysis techniques, implementing classifiers such as K-means, ANN, and Fuzzy to categorize children's reading and cognitive metrics.

Rekha, S., Gollapudi, S., Sampath, H., & Indurkhya, B. [15] The research paper introduces the Read-Aid tool, an assistive technology designed to improve reading skills in children with dyslexia by utilizing a masked-text technique.

III. METHODOLOGY/EXPERIMENTAL

1. Development of Dyslexia Screening Tests

Our project entails three dyslexia screening tests utilizing LLM based on their learning capabilities.

Phonological Awareness Test

- Multiple words are initially generated using DistilBERT's Text to Text model.
- The words are segregated into easy, medium and hard categories and stored in MySQL.
- The words are converted to audio using Google's Text to Speech API and are provided to the user.
- The screenshot of the whiteboard on which the word is written is taken and then the word is extracted using Optical Character Recognition.
- DistilBERT's Similarity Analysis is then done to generate a similarity score.

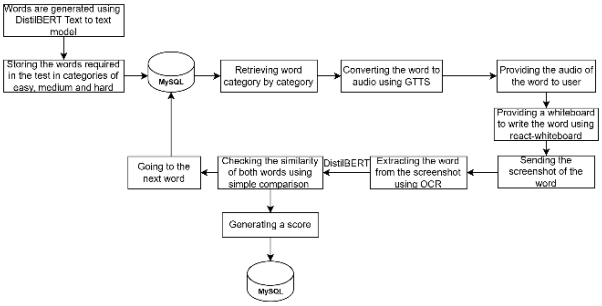


Fig 1. Phonological Awareness Test

Kauffman Assessment Battery Test

- Memory sequences are created using words, images and audios.
- Words - DistilBERT Text to Text model
- Images - Stable Diffusion Text to Image model
- Audio - DeepSpeech Text to Audio model.
- The sequences are provided to the user for a 10 second duration to memorize it.
- Score is based on the recreation of the same.

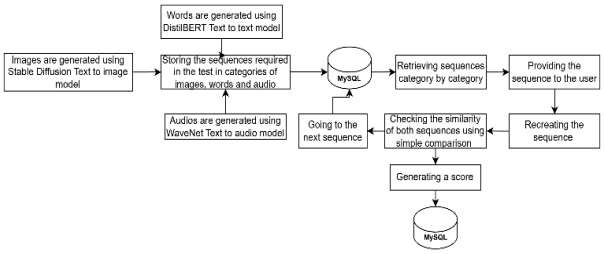


Fig 2. Kauffman Assessment Battery Test

Gray Oral Reading Test

- Passages are generated using DistilBERT's text-to-text model.
- Divided into easy, medium & hard categories.
- The passages are provided to user for reading.

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- The user audio is recorded using the MediaRecorder API.
- Fluency analysis is performed on the audio files using DeepSpeech audio to text model.
- The time taken, reading speed and a score based on the fluency is provided to the user.

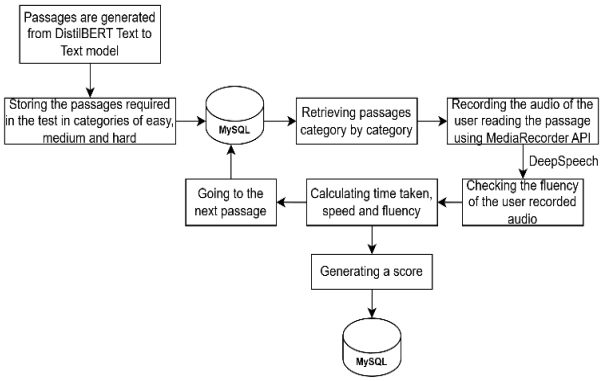


Fig 3. Gray Oral Reading Test

Score Generation

Dyslexic score is used to provide personalized learning paths which is calculated using a Random Forest model applied on the scores of the three tests.

2. Personalized Learning Path:

The personalized learning path is as follows:

Severity	Dyslexic Score	Feature Provided
Easy	<50%	Reading passages & comprehensions
Medium	50% - 75%	Reading passages, comprehensions & memory games
Hard	>75%	Reading passages, comprehensions, memory games & phonological improvement games

Table 1. Personalized Learning Paths

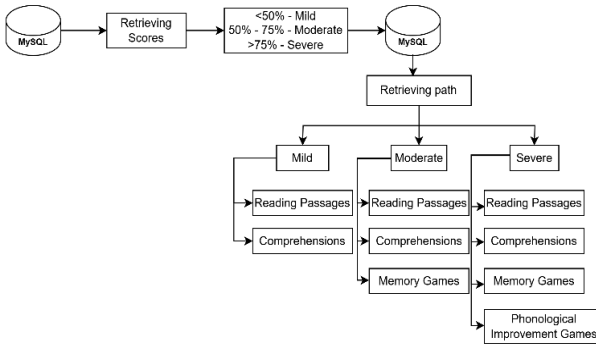


Fig 4. Personalized Learning Paths

Reading Passages

- Passages are generated using DistilBERT's model.
- Supporting audio for the passages are generated using WaveNet's Text to Audio model.
- The meaning of the words are provided to the users by the Dictionary API.
- The option to listen to each individual sentence in a multitude of speeds and voices is provided to the user using the WebSpeech API.
- User's audio is recorded by MediaRecorder API.
- DeepSpeech performs the fluency analysis.
- The time taken and reading speed are calculated.

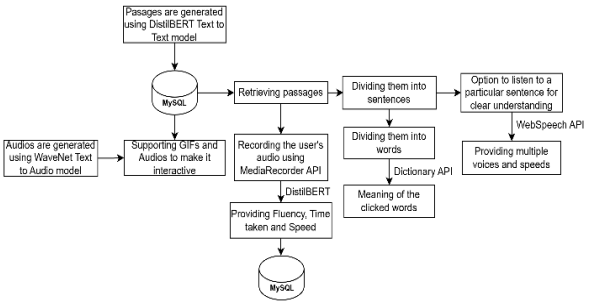


Fig 5. Reading Passages

Reading Comprehensions

- Custom Comprehensions are generated for the user using DistilBERT's Text to Text model. The comprehensions are then again sent as input to the LLM and suitable questions relevant for the comprehension are generated.
- Supporting Audio's generated using WaveNet's Text to Audio model as well as GIFs for the same are provided to the user.
- The answers to the questions are recorded and then compared to the answers generated by the DistilBERT model to generate a score for the same which is stored in the MySQL based database.

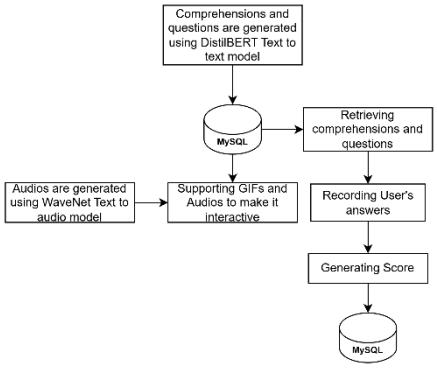


Fig 6. Reading Comprehensions

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Memory Games

- Images for the cards are generated using the Stable Diffusion Text to Image model.
- Each card is to be matched with its corresponding pair.

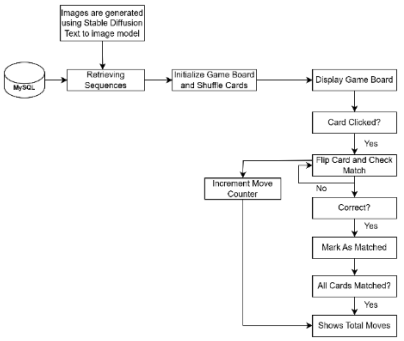


Fig 7. Memory Games

Phonological Improvement Assistant

- Audio is generated using WaveNet's Text to Audio model.
- Puzzle Grid consisting of a shuffled word is initialized and user needs to swap the tiles to create the word in the proper order.

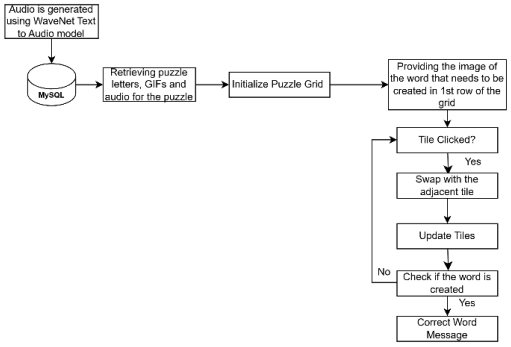


Fig 8. Phonological Improvement Assistant

3. AI Powered Document Simplification

- DistilBERT is applied on the text is extracted using PyPDF2 from the files uploaded.
- Further, DistilBERT identifies important words from the summarized text which are then higlighted using Dynamic CSS.
- Finally, summarized PDF is generated with key-words highlighted and entire text summarized.

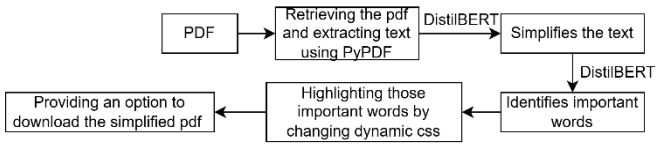


Fig 9. AI Powered Document Simplification

4. AI Driven Mind Map Generation

- DistilBERT is applied on the text is extracted using PyPDF2 from the files uploaded.
- Important points generated using DistilBERT are used to create a mind map using reactflow library.
- The model generates important words which are highlighted using dynamic CSS.

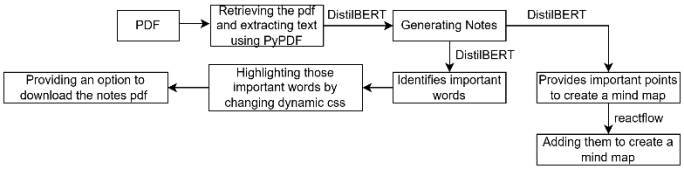


Fig 10. AI Powered Document Simplification

5. AI Powered Writing Assistant

- Identifies mistakes and improves the text.
- Text is corrected using fine-tuned DistilBERT.
- Image is corrected using fine-tuned OCR and DistilBERT on dyslexic handwriting styles.

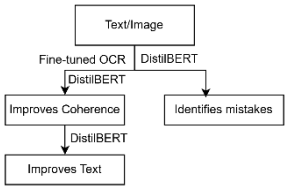


Fig 11. AI Powered Writing Assistant

6. AI Chatbot

- Input is in the form of Text, Audio and Images.

Input	Model used
Text	DistilBERT
Audio	DeepSpeech + DistilBERT
Image	OCR + DistilBERT

Table 2. LLM model used for various inputs

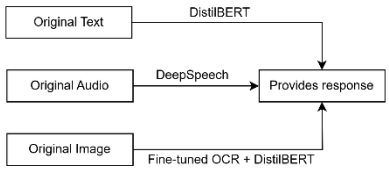


Fig 12. AI Chatbot

7. Integration of Frontend and Backend

- All the above features are integrated using react for the frontend and flask for the backend.
- Axios is used to connect the frontend to the backend.
- Multiple hugging face models are used in the course of this project to bring the above features to life.

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IV. Results and Discussions

The following section gives a detailed comparison of multiple models used in our system with other alternatives detailing the superior accuracy and performance achieved by our system.

1. Text-to-Text Models fine-tuned with dyslexic dataset

Models	Accuracy
BERT	82%
RoBERTa	80%
DistilBERT (Our System)	85%

Table 3. Comparison of Text-to-Text models

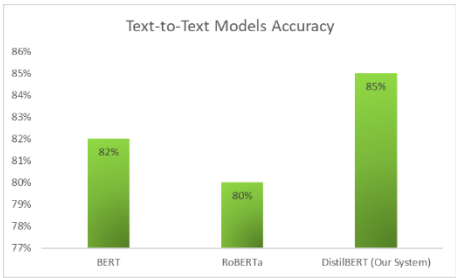


Fig 13. Accuracy comparison of Text-to-Text models

2. Text-to-Image models

Models	Accuracy
DALL-E 2	88%
Midjourney	85%
Stable Diffusion (Our System)	92%

Table 4. Comparison of Text-to-Image models

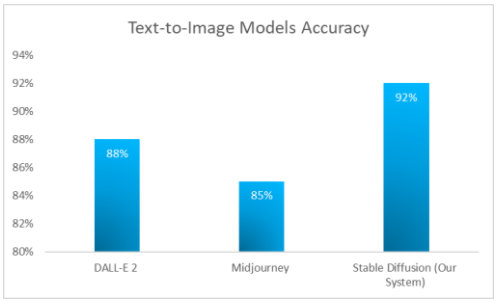


Fig 14. Accuracy comparison of Text-to-Image models

3. Audio-to-Text models

Models	Accuracy
Wav2Vec 2.0	88%
Whisper (OpenAI)	89%
DeepSpeech (Our System)	91%

Table 5. Comparison of Audio-to-Text models

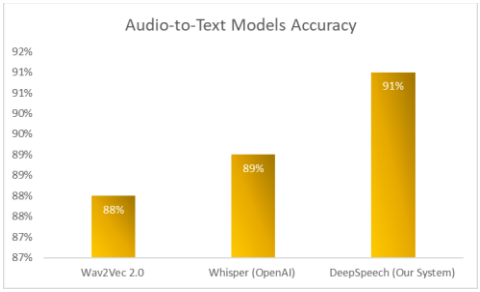


Fig 15. Accuracy comparison of Audio-to-Text models

4. Text-to-Audio models

Models	Accuracy
FastSpeech	87%
Tacotron 2	92%
WaveNet (Our System)	95%

Table 6. Comparison of Text-to-Audio models

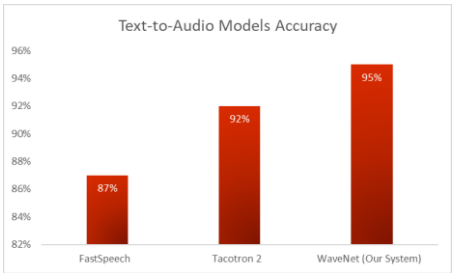


Fig 16. Accuracy comparison of Text-to-Audio models

V. Future Scope

The basic form of the proposed AI-powered system lays a solid foundation for future development not only in the field of personalized education but in its assistive technologies, too. In the future, its functions can be further enhanced to support multiple languages in the near future so as to help dyslexic persons all over the world. Further work with educational institutions, and participation of healthcare professionals, will also serve to refine the system's diagnostic and support capabilities.

VI. Conclusion

This research demonstrates the potential of AI-powered systems to revolutionize the assessment and support of individuals with dyslexia. The high-level machine learning model DistilBERT and DeepSpeech are incorporated into the system, which allows thorough dyslexia testing with personalized solutions for learning. Further incorporation of other tools, which include document simplification, mind map preparation, and writing assistance, has added to more usability and accessibility features that foster a supportive learning environment. These results confirm that the system is efficient for classifying severity with respect to dyslexia and providing appropriate interventions.

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